Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Digital Learning Handout

Part A: Content Design

Course Title	Deep Neural Networks
Course No(s)	AIML* ZG511
Credit Units	4
Credit Model	3-1-0
Course Author	Prof. Seetha Parameswaran
Version No:	1
Instructor(s)	Monali Mavani (Lead),S Bhagath, Manjula A K,Suresh Kumar,Vishali
Date:	21/02/2025

Course Description:
Introduction to neural networks, approximation properties, back propagation, deep network training, regularization and optimization, convolution neural networks, recurrent neural networks, attention models, transformers, neural architecture search, federated learning, meta learning, applications in time series modelling and forecasting, online (incremental) learning.

Course Objectives

No	Course Objective
CO1	Apply deep learning techniques to solve complex problems in various domains, demonstrating proficiency in designing and implementing neural network architectures including feedforward, convolutional, and recurrent networks.
CO2	Analyze the performance and behavior of deep learning models, including the impact of depth, optimization challenges, and regularization techniques, to enhance model efficiency and effectiveness.
CO3	Evaluate the strengths and limitations of different deep learning architectures and techniques for specific problem domains, considering factors such as computational complexity, data requirements, and model interpretability.
CO4	Design and implement innovative deep learning solutions that integrate multiple techniques, such as attention mechanisms and neural architecture search, to address challenging problems in areas like computer vision, natural language processing, and time series analysis.
CO5	Critically assess the latest advancements in deep learning, including transformers, federated learning, and meta-learning.

Text Book(s):

T1	Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). Dive into deep learning. Cambridge University Press.

Reference Book(s) & other resources:

R1	Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1, No. 2). Cambridge: MIT press.
R2	Eugene, C. (2019). Introduction to deep learning. MIT Press
R3	Chollet, F., & Chollet, F. (2021). Deep learning with Python. Simon and Schuster.
R4	Brownlee, J. (2018). Deep learning for time series forecasting: predict the future with MLPs, CNNs and LSTMs in Python. Machine Learning Mastery.
R5	Elsken, T., Metzen, J. H., & Hutter, F. (2019). Neural architecture search: A survey. Journal of Machine Learning Research, 20(55), 1-21.

Learning Outcomes: Students will be able to

LO1	Implement and optimize various deep neural network architectures (including feedforward, convolutional, and recurrent networks) to solve complex problems in areas such as image recognition, natural language processing, and time series forecasting.
LO2	Analyze the impact of different hyperparameters, optimization algorithms, and regularization techniques on the performance of deep learning models, and apply this knowledge to improve model training and generalization.
LO3	Evaluate the effectiveness of different deep learning architectures and techniques for specific problem domains, considering factors such as computational efficiency, data requirements, and model interpretability.
LO4	Design and implement advanced deep learning solutions that incorporate attention mechanisms, transformers, or neural architecture search techniques to address challenging real-world problems.

Prerequisites

AIML* ZC416 Mathematical Foundations for Machine Learning
AIML* ZC418 Introduction to Statistical Methods
AIML* ZG565 Machine Learning

Modular Content Structure

1. Fundamentals of Neural Network
    - Supervised, unsupervised, semi-supervised and reinforcement learning problems.
    - Why Deep Learning?
    - Applications of Deep Learning
    - Perceptron and Perceptron learning algorithm
    - Multilayer Perceptron (MLP)
    - MLP as classifiers and Universal approximators
    - Issue of Depth and Width

2. Deep Feedforward Neural Networks
    - Forward and backward propagation
    - Computation graph
    - Gradient Descent algorithm
    - Impact of depth in DNN
3. Optimization of Deep models
    - Challenges in Neural Network Optimization – saddle points and plateau
    - Non-convex optimization intuition
    - Overview of optimization algorithms
    - Mometum based algorithm
    - Algorithms with Adaptive Learning Rates
4. Regularization for Deep models
    - Model Selection
    - Underfitting, and Overfitting
    - L1 and L2 Regularization
    - Dropout
    - Challenges - Vanishing and Exploding Gradients, Covariance shift
    - Parameter Initialization
    - Batch Normalization
5. Convolutional Networks
    - Convolutions for Images
    - Learning a Kernel
    - Padding and stride, Channels, Pooling
    - Design of CNN
    - Popular CNN architectures
    - Transfer Learning
    - Applications of CNN
6. Sequence Models
    - Recurrent Neural Networks
    - Back-propagation through time
    - Challenge - Exploding - Vanishing gradient and Gates
    - Popular RNN architectures
    - Applications of RNNs
7. Attention Mechanism
    - Attention Pooling
    - Attention Scoring Functions
    - Multi-Head Attention, Self-Attention and Positional Encoding
    - Transformer architecture
    - Applications of Transformers
8. Neural Network search
    - Search Space
    - Search algorithms
    - Evaluation Strategy
9. Time series Modelling and Forecasting
    - Univariate, Multivariate and Multi-step CNN Models
    - Univariate, Multivariate and Multi-step LSTM Models

10. Other Learning Techniques
    - Federated learning
    - Meta learning
    - Online (incremental) learning

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	**Fundamentals of Neural Network**	• Supervised, unsupervised, semi-supervised and reinforcement learning problems. • Why Deep Learning? • Applications of Deep Learning • Biological neuron vs artificial neuron • Connectionism model • Perceptron • Perceptron learning algorithm • XOR Problem	T1 – Ch1 http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf
2	**Fundamentals of Neural Network**	• Perceptron and Perceptron learning algorithm • Multilayer Perceptron (MLP) • MLP as classifiers and Universal approximators • Issue of Depth and Width	http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter2_UniversalApproximators.pdf
3	**Deep Feedforward Neural Networks**	• Forward and backward propagation • Backward Propagation (review) • Training a DNN using Gradient Descent algorithm (review) • Computational Graphs (review as already discussed in MFML) • Activation Functions • Softmax Regression • Impact of depth in DNN	T1 – Ch4 and Ch3.4
4	**Optimization of Deep models**	• Challenges in Neural Network Optimization – saddle points and plateau • Non-convex optimization intuition • Stochastic Gradient Descent (SGD), Minibatch SGD • Overview of Rprop, Quickprop • Momentum, Nastrov's Accelarated Momentum • Algorithms with Adaptive Learning Rates,	T1 – Ch11

		Adagrad, RMSprop, ADAM	
5	**Regularization for Deep models**	• Model Selection • Underfitting, and Overfitting • L1 and L2 Regularization • Dropout • Challenges - Vanishing and Exploding Gradients, Covariance shift • Parameter Initialization • Batch Normalization	T1 – Ch4 , 7.5
6	**Convolutional Networks**	• Convolutions for Images • Learning a Kernel • Padding and stride, Channels, Pooling • Design of CNN	T1 – Ch6, Ch7
7	**Convolutional Networks**	• Popular CNN architectures • Transfer Learning • Applications of CNN	T1 – Ch6, Ch7
8	**Review**	Review of contact session 1 to 7	
9	**Sequence Models**	• Recurrent Neural Networks • Types of Sequences and RNNs • Back-propagation through time • Challenge - Exploding - Vanishing gradient and Gates	T1 – Ch8, Ch9
10	**Sequence Models**	• Popular RNN architectures - GRU, LSTM, BiLSTM • Applications of RNNs	T1 – Ch8, Ch9
11	**Attention Mechanism**	• Attention Pooling • Attention Scoring Functions • Multi-Head Attention, Self-Attention and Positional Encoding	T1 – Ch10
12	**Attention Mechanism**	• Transformer architecture • Applications of Transformers	T1 – Ch10
13	**Neural Network search**	• Search Space • Search algorithms • Evaluation Strategy	R5
14	**Time series Modelling and**	• Univariate, Multivariate and Multi-step CNN Models	R4 Ch-8, Ch-9

	Forecasting	• Univariate, Multivariate and Multi-step LSTM Models	
15	**Other Learning Techniques**	• Federated learning • Meta learning • Online (incremental) learning	https://arxiv.org/pdf/1902.04885.pdf https://arxiv.org/pdf/1903.10635.pdf https://arxiv.org/pdf/2004.05439.pdf https://arxiv.org/pdf/1802.07569.pdf https://arxiv.org/pdf/1904.07734v1.pdf
16	**Review**	Review of contact session 9 to 15	

Experiential Learning Components:

1. Lab work: 7
2. Project work: 0
3. Case Study: 0
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of

a) MLP
b) CNN
c) RNN
d) Transformer

Scope of Experiential Learning Component:

Programming language: Python
Tools and libraries: Jupyter, Numpy, Scipy, Pandas, ScikitLearn, Matplotlib, Seaborn, Tensorflow, Pytorch, Keras

Lab Infrastructure:

Online / Open source / Virtual lab / Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Introduction to Tensorflow, Keras	2
2	Computational graph in Pytorch	3

3	Deep Neural Network with Back-propagation and optimization	4
4	CNN	6
5	RNN, LSTM	9
6	Transformers	12
7	Time series forecasting	15

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Quiz	Online	10%	1 week	March 1-10, 2026
	Assignment/Lab Assignment / Lab Exams	Online	20 %	10 days	April 1-10, 2026
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	14/03/2026 (FN)
EC - 3	Comprehensive Exam	Open Book	40%	2 ½ Hours	23/05/2026 (FN)

EC1* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: https://elearn.bits-pilani.ac.in

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: "open book" means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.
   EC3 Guidelines
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

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